

Embedded Systems mit RISC-V

2. Hallo, Welt!

V1.0, Patrick Ritschel

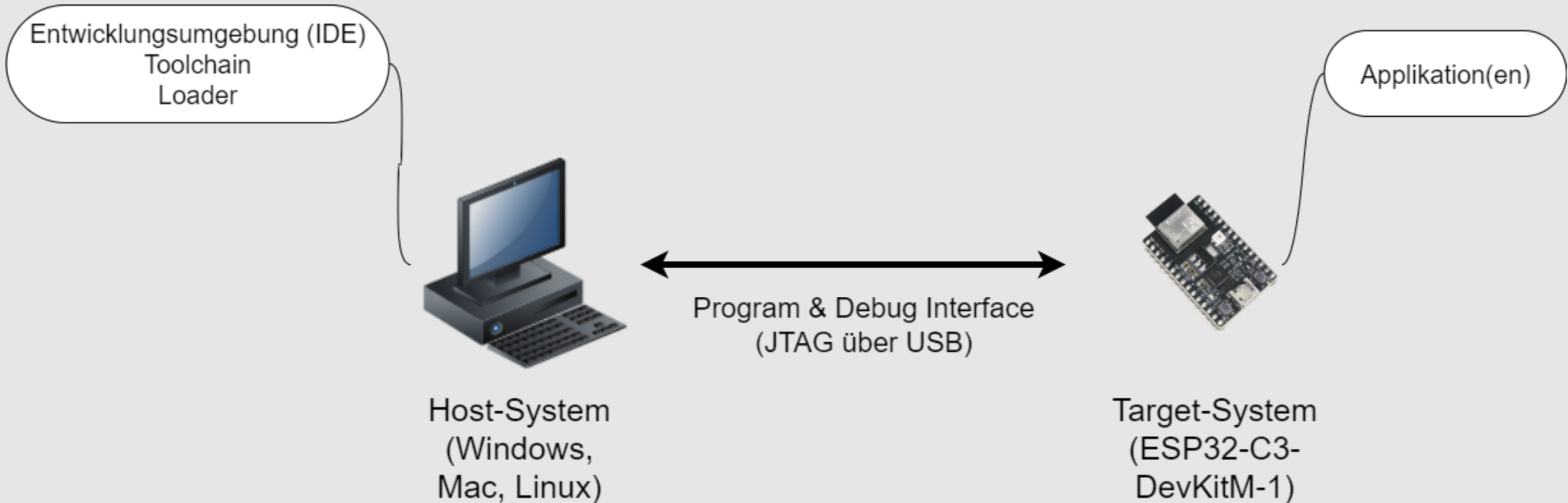


IIRFilter

Filter

```
float iirfilter_filterValue(Filter* pFilter, float value) {  
    float y = 0.0f;  
    for (int i = 0; i < pFilter->order; i++)  
        y += (pFilter->coeffs[i] * value);  
    return y;  
}
```

Cross-Platform-Entwicklung Komponenten

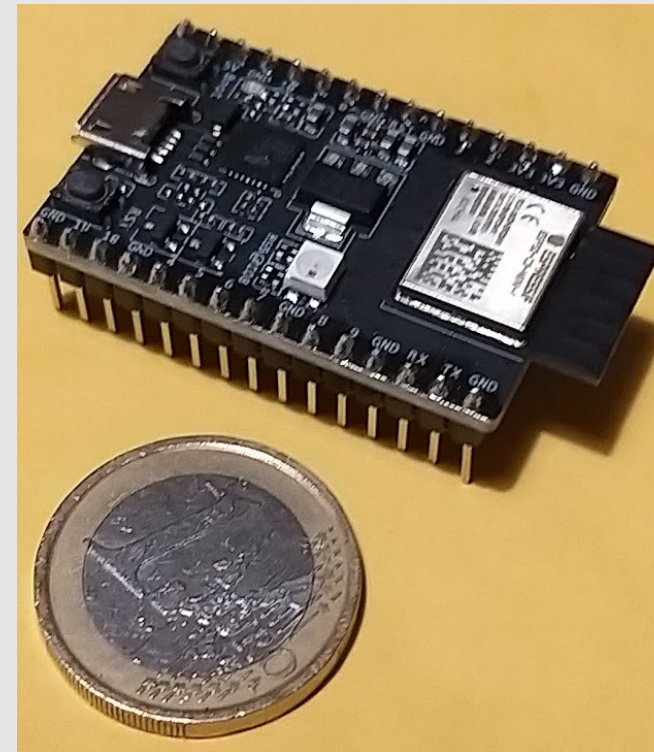
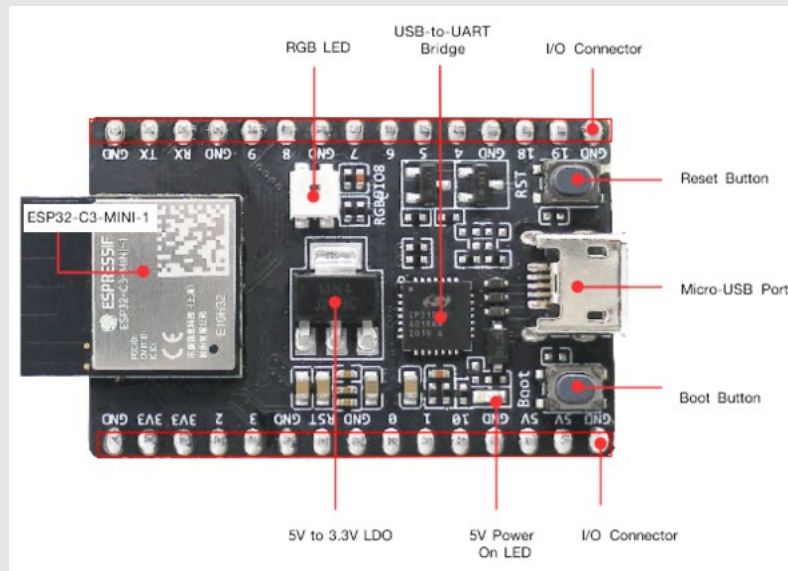


IIRFilter

Filter

```
float iirfilter_filterValue(Filter* pFilter, float value) {
```

Development Board ESP32-C3-DevKitM-1

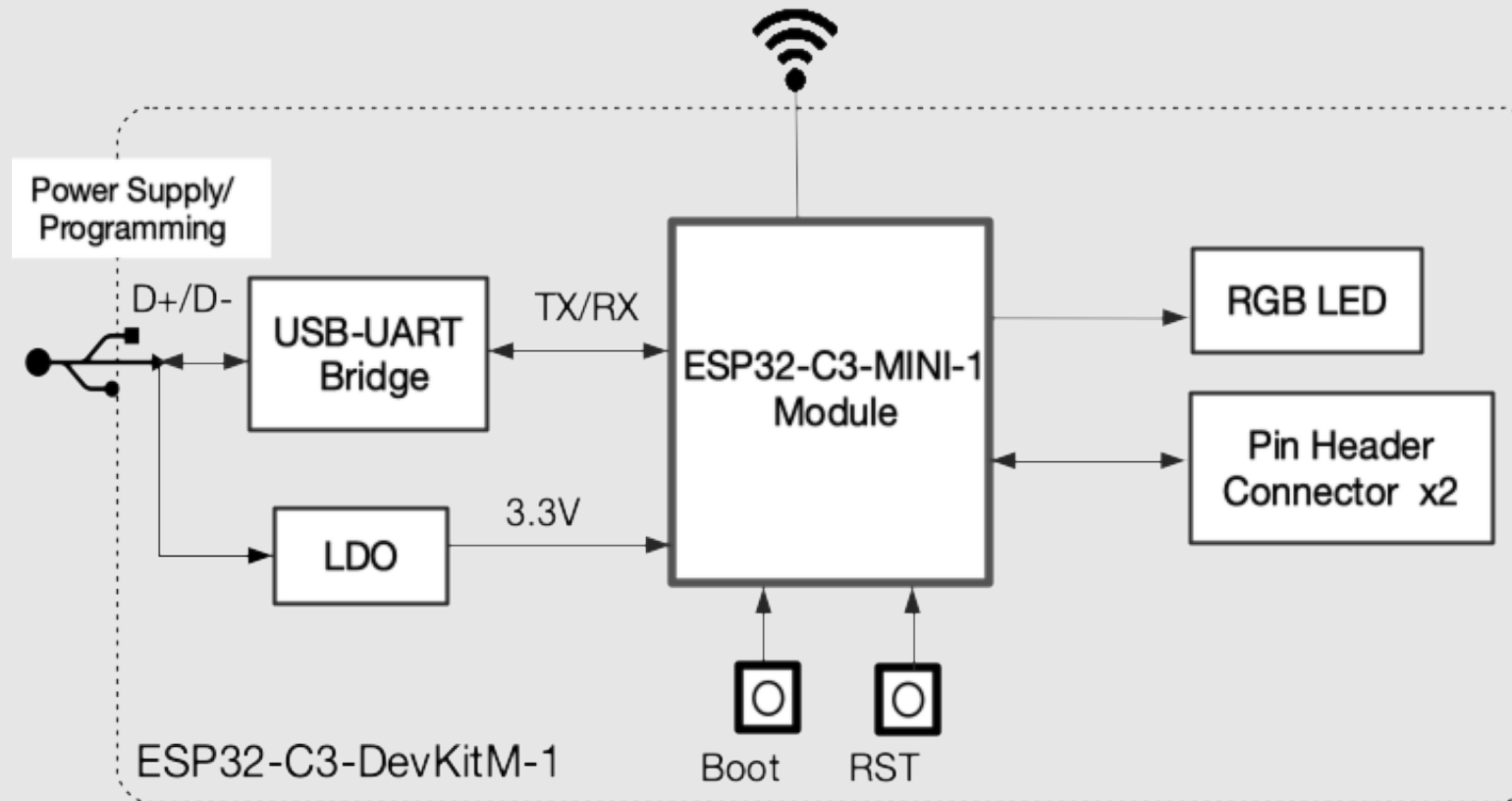


IIRFilter

Filter

```
float iirfilter_filterValue(Filter* pFilter, float value) {
```

Development Board Schematischer Aufbau

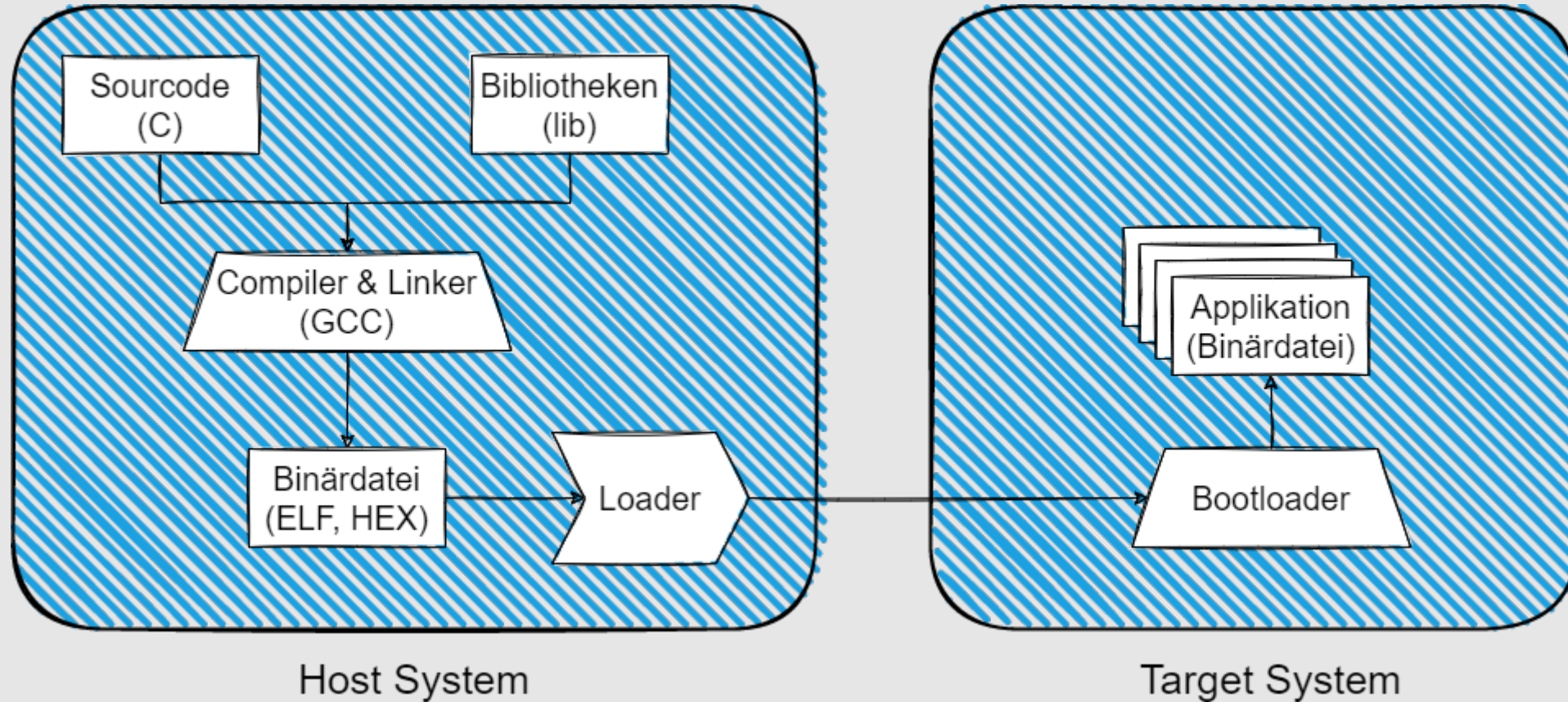


IIRFilter

Filter

```
float iirfilter_filterValue(Filter* pFilter, float value) {
```

Cross-Platform Development Toolchain



IIRFilter

Filter

```
float iirfilter_filterValue(Filter* pFilter, float value) {
```


Eclipse IDE Integrated Development Environment



esp-workspace5 - hello_world/main/hello_world_main.c - Espressif-IDE

File Edit Source Refactor Navigate Search Project Run Espressif Window Help

Run hello_world on: esp32c3

Project Explorer Console

<terminated> hello_world [ESP-IDF Application] C:\Users\prits\espressif\python_env\idf5.0_py3.8_env\Scripts\python.exe (12393 compressed) at 0x00000000 in 0.4 seconds (effective 370.2 kbi)

Hash of data verified.

Compressed 155776 bytes to 84450...

Writing at 0x00010000... (16 %)

Writing at 0x00019bf6... (33 %)

Writing at 0x000205cc... (50 %)

Writing at 0x0002764a... (66 %)

Writing at 0x0002dd08... (83 %)

Writing at 0x00034da0... (100 %)

Wrote 155776 bytes (84450 compressed) at 0x00010000 in 2.3 seconds (effective 535.8 kbi)

Hash of data verified.

Compressed 3072 bytes to 103...

Writing at 0x00008000... (100 %)

Wrote 3072 bytes (103 compressed) at 0x00008000 in 0.1 seconds (effective 401.1 kbit/s)

Hash of data verified.

Leaving...

Hard resetting via RTS pin...

Done

Connected - Encoding: windows-1252

hello_world_main.c

```
1 #include <stdio.h>
2
3 void app_main(void)
4 {
5     printf("Hello world!\n");
6 }
7
```

Outl x Build

stdio.h

app_main(void) : void

Problems Tasks Properties Terminal

COM4 x

```
I (103) esp_image: segment 2: paddr=00018478 vaddr=40380000 size=07ba0h ( 31648) 1
oad
I (116) esp_image: segment 3: paddr=00020020 vaddr=42000020 size=1332ch ( 78636) m
ap
I (131) esp_image: segment 4: paddr=00033354 vaddr=40387ba0 size=02cf0h ( 11504) 1
oad
I (134) esp_image: segment 5: paddr=0003604c vaddr=50000010 size=00010h ( 16) 1
oad
I (140) boot: Loaded app from partition at offset 0x10000
I (143) boot: Disabling RNG early entropy source...
I (360) cpu_start: Pro cpu up.
I (369) cpu_start: Pro cpu start user code
I (369) cpu_start: cpu freq: 16000000
I (369) cpu_start: Application information:
I (372) cpu_start: Project name: hello_world
I (377) cpu_start: App version: 1.0.0
I (382) cpu_start: Compile time: Nov 11 2022 13:32:48
I (388) cpu_start: ELF file SHA256: ee206b9c56808dbd...
I (394) cpu_start: ESP-IDF: v5.0-beta1-827-g166effd4b8
I (400) heap_init: Initializing. RAM available for dynamic allocation:
I (408) heap_init: At 3FC8CD40 len 0004F9D0 (318 KiB): DRAM
I (414) heap_init: At 3FCDC710 len 00002950 (10 KiB): STACK/DRAM
I (421) heap_init: At 50000020 len 00001FE0 (7 KiB): RTCRAM
I (428) spi_flash: detected chip: generic
I (432) spi_flash: flash io: dio
W (436) spi_flash: Detected size(4096k) larger than the size in the binary image h
eader(2048k). Using the size in the binary image header.
I (449) cpu_start: Starting scheduler.
Hello world!
```

Ein erstes Programm

hello_world-Beispiel des ESP-IDF



```
#include <stdio.h>
/* [...] other */
void app_main(void) {
    printf("Hello world!\n");

    /* [...] print system infos and reset system after 10s */
}
```

IIRFilter

Filter

```
float iirfilter_filterValue(Filter* pFilter, float value) {
```

Eclipse IDE Debugging



Debug hello_world Debug

hello_world_main.c

```
1 #include <stdio.h>
2
3 void app_main(void)
4 {
5     printf("Hello World!\n");
6 }
```

hello_world Debug [ESP-IDF GDB Open]

(3268) hpmcounter3h (/32)
(3269) hpmcounter4h (/32)
(3270) hpmcounter5h (/32)

IIRFilter

Filter

```
float iirfilter_filterValue(Filter* pFilter, float value) {
```